

Statistics MCQ Question Bank

Second Paper

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1 Introduction to Probability

1.1 Permutation-Combination

1. Three objects can be placed in 2 positions in – ways.
(a) 3 (b) 4 (c) 6 (d) 8
2. In how many ways can a team of 2 be formed from 4 people?
(a) 4 (b) 6 (c) 8 (d) 12
3. ${}^n p_r =$
(a) $\frac{n!}{(n-r)!}$ (b) $\frac{n!}{(n+r)!}$ (c) $\frac{n!}{r!}$ (d) $\frac{n!}{(r-n)!}$
4. ${}^n C_r =$
(a) $\frac{n!}{(n-1)!(n+r)!}$ (b) $\frac{r!}{n!(n-r)!}$ (c) $\frac{n!(n-1)!}{r!}$ (d) $\frac{n!}{(r-n)!}$

1.2 Conceptual Questions

5. What is the probability that at least one item in a sample space will occur?
(a) 0 (b) 0.5 (c) 1 (d) Undefined
6. The probability of two disjoint sets happening together is:
(a) 0.5 (b) 0 (c) 1 (d) $0 \leq x < 1$
7. How many additive laws of probability are there?
(a) 1 (b) 2 (c) 3 (d) 4
8. $P(A \cup B) = P(A) + P(B)$ implies A & B are –
(a) Disjoint (b) Independent (c) Joint (d) Independent
9. $P(A \cap B) = P(A) \times P(B)$ implies A & B are –
(a) Disjoint (b) Independent (c) Joint (d) Independent
10. Which is the formula of classical approach of probability?
(a) $P = \frac{\text{No. of favorable outcomes}}{\text{Total no. of possible outcomes}}$ (b) $P = \frac{\text{No. of total outcomes}}{\text{No. of favorable outcomes}}$
(c) $P = \lim_{n(S) \rightarrow \infty} \frac{n(A)}{n(S)}$ (d) $P = \lim_{n(A) \rightarrow \infty} \frac{n(A)}{n(S)}$
11. Which is the formula of empirical/relative frequency approach of probability?
(a) $P = \frac{\text{No. of favorable outcomes}}{\text{Total no. of possible outcomes}}$ (b) $P = \frac{\text{No. of total outcomes}}{\text{No. of favorable outcomes}}$
(c) $P = \lim_{n(S) \rightarrow \infty} \frac{n(A)}{n(S)}$ (d) $P = \lim_{n(A) \rightarrow \infty} \frac{n(A)}{n(S)}$
12. What is the correct formula for conditional probability?
(a) $P(A|B) = \frac{P(A \cap B)}{P(B|A)}$ (b) $P(A|B) = \frac{P(A \cap B)}{P(A)}$ (c) $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (d) $P(A|B) = \frac{P(B|A)}{P(B|A)}$
13. The third axiom of probability is –
(a) $0 \leq P(A) \leq 1$ (b) $P(S) = 1$
(c) $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^{\infty} P(A_i)$ (d) $P(A) = 1 - P(A)$

14. **Possible value of probability**
 i. -1 ii. 0.5 iii. 0
Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
15. **An act repeated under some specific conditions is called –**
 (a) Event (b) Experiment (c) Sample (d) Sample space
16. **$P(0)$ implies –**
 (a) A certain event (b) An uncertain event (c) An impossible event (d) A probable event
17. **Events having some common elements are called –**
 (a) Complementary events (b) Mutually exclusive events
 (c) Exhaustive events (d) Non-Mutually exclusive events events
18. **The minimum value of probability is**
 (a) $-\alpha$ (b) 1 (c) 0 (d) -1
19. **Each element of sample space is called–**
 (a) Trial (b) Experiment (c) Variable (d) Sample Point
20. **Two events not occurring together are called–**
 (a) dependent Events (b) Independent Events
 (c) Mutually Exclusive Events (d) Marginal Events
21. **If A and B are independent, which formula is correct?**
 (a) $P(A \cap B) = P(A) \cdot P(B)$ (b) $P(A \cap B) = P(\bar{A}) \cdot P(B)$
 (c) $P(A \cap B) = P(A) \cdot P(\bar{B})$ (d) $P(A \cap \bar{B}) = P(A) \cdot P(B)$
22. **Which of the following are disjoint events?**
 (a) $A = \{1, 2, 3\}, B = \{4, 5\}$ (b) $A = \{a, b\}, B = \{b, c\}$
 (c) $A = \{0\}, B = \{0, 1\}$ (d) $A = \{x, y\}, B = \{x, y\}$
23. **Which of the following are disjoint events?**
 (a) $P = \{1, 2\}, Q = \{2, 3\}$ (b) $P = \{x\}, Q = \{x, y\}$
 (c) $P = \{1, 3\}, Q = \{3, 5\}$ (d) $P = \{m, n\}, Q = \{p, q\}$
24. **Let the sample space be $S = \{1, 2, 3, \dots, 10\}$. Which of the following pairs of events are disjoint?**
 i. A: Number is prime, B: Number is greater than 3
 ii. A: Number is even, B: Number is divisible by 3
 iii. A: Number is less than 5, B: Number is greater than 6
Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
25. **Let $S = \{1, 2, \dots, 10\}$. Which of the following event pairs are disjoint?**
 i. A: Number is divisible by 2, B: Number is divisible by 5
 ii. A: Number is less than 7, B: Number is odd
 iii. A: Number is a prime, B: Number is a multiple of 4
Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

26. Let the sample space be $S = \{1, 2, 3, \dots, 10\}$. Which of the following pairs of events are disjoint?

- i. A : Number is a multiple of 4, B : Number is odd
- ii. A : Number is less than 4, B : Number is greater than 8
- iii. A : Number is a square, B : Number is even

Which one is correct?

- (a) i and ii
- (b) i and iii
- (c) ii and iii
- (d) i, ii and iii

27. Let $S = \{1, 2, 3, \dots, 10\}$. Which of the following pairs of events are disjoint?

- (a) A : Multiples of 3, B : Multiples of 5
- (b) A : Prime numbers, B : Even numbers greater than 2
- (c) A : Numbers less than 4, B : Numbers greater than 6
- (d) All of the above

1.3 Numbers

28. A number is randomly chosen from a list of 10 consecutive positive integers. What is the probability that the number selected is greater than the average (arithmetic mean) of all 10 integers?

- (a) $\frac{1}{3}$
- (b) $\frac{3}{4}$
- (c) $\frac{4}{10}$
- (d) $\frac{1}{2}$

29. 10 out of each 100 people in a city walk to the office. If one is picked randomly, what is the probability s/he does not walk to the office?

- (a) 0.95
- (b) 0.10
- (c) 0.90
- (d) 0.01

30. In a school, 15 out of 100 students prefer online classes over in-person classes. If a student is selected randomly, what is the probability that they prefer in-person classes?

- (a) 0.15
- (b) 0.85
- (c) 0.75
- (d) 0.25

31. A factory reports that 8 out of every 100 manufactured items are defective. If an item is chosen at random, what is the probability that it is not defective?

- (a) 0.08
- (b) 0.92
- (c) 0.80
- (d) 0.12

32. A hospital study finds that 12% of patients do not prefer evening appointments. If a patient is selected at random, what is the probability that they prefer evening appointments?

- (a) 0.12
- (b) 0.78
- (c) 0.88
- (d) 0.18

33. A survey shows that 5 out of every 200 customers in a store pay with cash. If a customer is picked randomly, what is the probability that they pay using another method?

- (a) 0.050
- (b) 0.500
- (c) 0.975
- (d) 0.025

1.4 Coin-Die

34. Tossing a die r times generates how many outcomes?

- (a) $6 \times r$
- (b) r^6
- (c) 6^r
- (d) 2^r

35. Tossing a coin r times generates how many outcomes?

- (a) $2 \times r$
- (b) r^2
- (c) 2^r
- (d) 6^r

36. A coin is thrown thrice. How many outcomes are generated?
 (a) 3 (b) 4 (c) 8 (d) 9
37. A coin is thrown twice. What is the probability of getting 2 heads?
 (a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{4}$
38. A fair coin is tossed twice. What is the probability of getting at least one tail?
 (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$
39. Two fair coins are tossed simultaneously. What is the probability of getting exactly one head?
 (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$
40. A coin is flipped twice. What is the probability of getting heads first and tails second?
 (a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{4}$
41. If two fair coins are tossed together, what is the probability of getting at least one head?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{3}{4}$ (d) $\frac{1}{4}$
42. A fair coin is tossed twice. What is the probability of getting two tails?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{2}{4}$
43. Two fair coins are tossed. What is the probability that at least one of them lands on tails?
 (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$
44. A die is thrown twice. This is called –
 (a) An experiment (b) sample space (c) A random experiment (d) A trial
45. If a neutral die is thrown, the probability of having a digit greater than 6 is
 (a) $\frac{1}{6}$ (b) $\frac{0}{6}$ (c) $\frac{2}{3}$ (d) $\frac{3}{6}$
46. Tossing a coin twice generates how many outcomes?
 (a) 4 (b) 16 (c) 8 (d) 2
47. A die is rolled twice. How many possible outcomes are there?
 (a) 6 (b) 12 (c) 36 (d) 18

1.5 Balls-Cards

48. There are 3 red, 4 black, and 5 white balls in an urn. If two balls are randomly taken, what is the probability that both are red?
 (a) $\frac{1}{66}$ (b) $\frac{1}{22}$ (c) $\frac{2}{22}$ (d) $\frac{3}{11}$
49. There are 3 red, 4 black, and 5 white balls in an urn. If two balls are randomly taken, what is the probability that neither is red?
 (a) $\frac{5}{11}$ (b) $\frac{6}{11}$ (c) $\frac{3}{11}$ (d) $\frac{5}{22}$
50. A jar contains 6 blue and 4 green marbles. If two marbles are drawn at random, what is the probability that both are blue?
 (a) $\frac{5}{18}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

51. A box has 7 black and 5 white balls. If one ball is drawn at random, what is the probability that it is not black?

- (a) $\frac{7}{12}$ (b) $\frac{5}{12}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

52. A bag contains 8 red and 6 white balls. If two balls are drawn at random, what is the probability that they are of different colors?

- (a) $\frac{24}{91}$ (b) $\frac{58}{91}$ (c) $\frac{48}{91}$ (d) $\frac{72}{91}$

53. A box contains 9 blue and 3 red balls. If two balls are randomly picked, what is the probability that at least one is red?

- (a) $\frac{3}{11}$ (b) $\frac{1}{3}$ (c) $\frac{18}{33}$ (d) $\frac{5}{11}$

Answer the next THREE questions based on the following information.

A card is drawn from of pack of playing cards.

54. What is the probability that the card is a King?

- (a) 0.0192 (b) 0.25 (c) 0.5 (d) 0.0769

55. P(The card is not from Diamonds)–

- (a) $\frac{1}{2}$ (b) 0 (c) $\frac{3}{4}$ (d) $\frac{1}{4}$

56. P(The card is red or Clubs)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Answer the next THREE questions based on the following information.

A card is drawn at random from a standard deck of 52 playing cards.

57. What is the probability that the card is a Queen?

- (a) 0.0192 (b) 0.0769 (c) 0.25 (d) 0.5

58. P(The card is not a Spade)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$

59. P(The card is black or a Heart)

- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Answer the next TWO questions based on the following information.

An urn contains 5 red, 7 blue, and 8 green balls.

60. What is the probability that the ball drawn is red?

- (a) 0.26 (b) 0.25 (c) 0.2 (d) 0.4

61. P(The ball drawn is not blue)–

- (a) $\frac{13}{20}$ (b) 0.5 (c) $\frac{7}{20}$ (d) $\frac{8}{20}$

1.6 Set-Problems

62. In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected. What is the probability that the student passed in at least one subject

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{4}{5}$

63. In a survey of 60 people, 35 liked coffee, 28 liked tea, and 12 liked neither. If a person is selected at random, what is the probability that they like at least one of the two beverages?

(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Answer the next two questions based on the following information

In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected.

64. What is the probability that the student passed in at least one subject?

(a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{4}{5}$

65. What is the probability that the student passed in both the subjects?

(a) $\frac{11}{20}$ (b) $\frac{9}{20}$ (c) $\frac{4}{5}$ (d) $\frac{19}{20}$

66. What is the probability that the student passed in passed in Mathematics alone?

(a) $\frac{1}{8}$ (b) $\frac{9}{20}$ (c) $\frac{3}{10}$ (d) $\frac{6}{13}$

67. For two independent events A and B , which one is correct?

(a) $P(A \cap B) = P(A) \times P(B)$ (b) $P(A \cup B) = P(A) + P(B)$
(c) $P(A \cap B) = P(A) - P(B)$ (d) $P(A \cup B) = P(A) \times P(B)$

68. For two mutually exclusive events A and B , which one is correct?

(a) $P(A \cap B) = P(A) \times P(B)$ (b) $P(A \cup B) = P(A) + P(B)$
(c) $P(A \cap B) = P(A) - P(B)$ (d) $P(A \cup B) = P(A) \times P(B)$

69. Which of the following correct?

(a) $\frac{P(A)}{P(B)} = \frac{P(B|A)}{P(A|B)}$ (b) $\frac{P(A)}{P(A|B)} = \frac{P(B|A)}{P(B)}$ (c) $\frac{P(A)}{P(B)} = \frac{P(B|A)}{P(B)}$ (d) $\frac{P(A)}{P(B)} = \frac{P(A|B)}{P(B|A)}$

70. The probability of rain is $\frac{1}{6}$ for any given day next week. What is the probability that it will rain on both Monday and Tuesday?

(a) $\frac{1}{6}$ (b) $\frac{1}{36}$ (c) $\frac{5}{6}$ (d) $\frac{1}{17}$

71. Given $P(A \cup B) = 0.7$, $P(A \cap B) = 0.2$, what are $P(A)$ and $P(B)$?

(a) $P(A) = 0.5$ and $P(B) = 0.4$ (b) $P(A) = 0.4$ and $P(B) = 0.6$
(c) $P(A) = 0.4$ and $P(B) = 0.3$ (d) $P(A) = 0.7$ and $P(B) = 0.3$

72. If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cup B) = 0.7$, what is $P(A \cap B)$?

(a) 0.2 (b) 0.1 (c) 0.3 (d) 0.4

73. Given $P(A) = 0.3$, $P(A \cup B) = 0.6$, and $P(A \cap B) = 0.1$, what is $P(B)$?

(a) 0.6 (b) 0.4 (c) 0.3 (d) 0.2

74. If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cap B) = 0.3$, what is $P(A \cup B)$?

(a) 0.8 (b) 0.9 (c) 0.7 (d) 1

75. If $P(A) = 0.2$, $P(B) = 0.3$, and $P(A \cup B) = 0.4$, what is $P(A \cap B)$?

(a) 0.9 (b) 0.2 (c) 0.3 (d) 0.1

76. Given $P(A) = 0.7$, $P(A \cup B) = 0.9$, and $P(A \cap B) = 0.5$, what is $P(B)$?

(a) 0.8 (b) 0.6 (c) 0.7 (d) 0.5

Answer the next two questions based on the following information

For two exhaustive events A & B, $P(A) = 0.7$ and $P(B) = 0.4$

77. $P(A \cap B) = ?$

- (a) 0.1 (b) 0.3 (c) 0.6 (d) 1

78. The events A & B are –

- i. independent
ii. dependent
iii. not mutually exclusive

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

Answer the next three questions using the following information

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{2} \text{ \& } P(A \cup B) = \frac{7}{12}$$

79. $P(A \cap B) = ?$

- (a) $\frac{5}{12}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{15}{16}$

80. $P(A \cap \bar{B}) = ?$

- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{5}{6}$ (d) $\frac{1}{12}$

81. What is the probability that B occurs or A does not occur?

- (a) $\frac{3}{4}$ (b) $\frac{7}{12}$ (c) $\frac{5}{12}$ (d) $\frac{11}{12}$

Answer the next three questions using the following information

$$P(C) = \frac{2}{5}, P(D) = \frac{3}{4} \text{ \& } P(C \cup D) = \frac{9}{10}$$

82. $P(C \cap D) = ?$

- (a) $\frac{1}{10}$ (b) $\frac{1}{4}$ (c) $\frac{7}{20}$ (d) $\frac{4}{5}$

83. $P(C \cap \bar{D}) = ?$

- (a) $\frac{1}{10}$ (b) $\frac{2}{5}$ (c) $\frac{3}{20}$ (d) $\frac{3}{10}$

84. What is the probability that D occurs or C does not occur?

- (a) $\frac{17}{20}$ (b) $\frac{7}{10}$ (c) $\frac{3}{4}$ (d) $\frac{11}{20}$

Answer the next three questions using the following information:

$$P(E) = \frac{1}{3}, P(F) = \frac{1}{4} \text{ \& } P(E \cap F) = \frac{1}{10}$$

85. $P(E \cup F) = ?$

- (a) $\frac{1}{58}$ (b) $\frac{3}{10}$ (c) $\frac{58}{60}$ (d) $\frac{58}{120}$

86. $P(E \cap \bar{F}) = ?$

- (a) $\frac{7}{40}$ (b) $\frac{7}{30}$ (c) $\frac{3}{10}$ (d) $\frac{1}{30}$

87. What is the probability that F occurs or E does not occur?

- (a) $\frac{11}{30}$ (b) $\frac{19}{30}$ (c) $\frac{13}{40}$ (d) $\frac{23}{30}$

88. An urn contains 10 red and 5 black balls. Two balls are drawn; what is the probability of getting two red balls?

- (a) $\frac{3}{7}$ (b) $\frac{4}{7}$ (c) $\frac{20}{21}$ (d) $\frac{2}{21}$

1.7 Situation Set

Answer the next two questions based on the following information

For two comprehensive events A and B , $P(A) = 0.8$, and $P(B) = 0.6$;

89. What is the value of $P(A \cap B)$?

- (a) 0.1 (b) 0.2 (c) 0.3 (d) 0.4

90. The events A and B are –

- i. independent
ii. dependent
iii. non-disjoint

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

1.8 Multiple Completion

91. $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{5}$, and $P(A \cap B) = \frac{1}{6}$

- i. A and B are independent
ii. $P(A \cup B) = \frac{7}{10}$
iii. $P(\bar{A}) = \frac{1}{3}$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

92. $P(A) = 0$ implies

- i. A is an impossible event
ii. A would occur in extreme cases
iii. $P(\bar{A})$ is a certain event

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

93. If A is an uncertain event, which one is possible?

- i. $0 < P(A) < 1$
ii. $P(A) = 0.1$
iii. $P(A) = 0$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

94. If a die is thrown once, the probability of getting even numbers is –

- i. A certain event
ii. A composite event
iii. An uncertain event

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

2 Random Variables

2.1 Concept of Random Variable

95. Which is a discrete random variable?

- (a) Age of students
- (b) Amount of Production in a factory
- (c) Height of workers
- (d) Page size in word processing softwares

96. A set of sample points tabulated along with their respective probabilities is an example of –

- (a) Probability distribution
- (b) Probability function
- (c) Frequency distribution
- (d) Marginal probability distribution

97. How many conditions does a probability density function have?

- (a) 2
- (b) 3
- (c) 4
- (d) 5

98. A coin is tossed twice and no. of heads appeared is denoted by X . How many possible values of X are there?

- (a) 1
- (b) 2
- (c) 0
- (d) 3

99. A die is thrown thrice and the number of times a 6 appears is denoted by X . How many possible values can X take?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

100. Which one is a property of marginal probability density function?

- (a) $\int_x f(x^2) dx = 1$
- (b) $\int_x f(x^2) dx = 0.5$
- (c) $\int_x f(x) dx = 1$
- (d) $P(x \geq 1)$

101. Which one is NOT an example of a continuous random variable –

- (a) Weight
- (b) Height
- (c) Time
- (d) Size of television

102. Integrated value of $\frac{1}{4}x^4$ –

- (a) $\frac{1}{20}x^5$
- (b) $\frac{1}{20}x^5 + c$
- (c) $\frac{1}{5}x^4$
- (d) $\frac{5}{4}x^5$

103. The conditions of a probability distribution are–

- i. $\sum P(X) = 1$
- ii. $\sum P(X) = 0$
- iii. $0 \leq P(X) \leq 1$

Which one is correct?

- (a) i and ii
- (b) i and iii
- (c) ii and iii
- (d) i, ii and iii

104. The conditions for a cumulative distribution function (CDF) are–

- i. $F(x)$ is non-decreasing.
- ii. $0 \leq F(x) \leq 1$
- iii. $\lim_{x \rightarrow \infty} F(x) = 1$

Which one is correct?

- (a) i and ii
- (b) ii and iii
- (c) i and iii
- (d) i, ii, and iii

105. **The properties of a discrete probability distribution table are—**
- $\sum P(X) = 1$
 - $P(X) \geq 0$ for all X
 - Each probability corresponds to a discrete value.
- Which one is correct?**
- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii, and iii
106. **What is $F(\infty)$ for a distribution function $F(x)$?**
- (a) $-\infty$ (b) -1 (c) 0 (d) 1
107. **What is $F(-\infty)$ for a distribution function $F(x)$?**
- (a) $-\infty$ (b) -1 (c) 0 (d) 1
108. **How many types of random variables are there?**
- (a) 2 (b) 3 (c) 4 (d) 5
109. **Which of the following is not a discrete random variable?**
- (a) number of students (b) Weight
(c) Number of heads in coin toss (d) Population
110. **Which one is a property of a probability distribution?**
- (a) $P(x_i) = 0$ (b) $P(x_i \neq 1)$ (c) $\sum P(x_i) = 1$ (d) $\int_x P(X)dx \leq 1$
111. **Which one is not a discrete random variable?**
- (a) Summation two die throw outcome (b) Weight
(c) Number of heads in five coin tosses (d) Released version number of a software
112. **Which one is not a discrete random variable?**
- (a) Number of students in a class (b) Weight of a package
(c) Shoe size (d) Total goals scored in a match
113. **Which variable type can skip certain whole numbers?**
- (a) Number of chapters read in a day (b) Weight of a person
(c) Number of floors in a building (d) Number of people boarding a train
114. **Which one is an example of a discrete random variable?**
- (a) The amount of liquid in a glass (b) Temperature readings at noon
(c) Number of defective items in a batch (d) Exact age in years
115. **Identify which one is not a discrete variable.**
- (a) Number of cookies eaten (b) Height of students
(c) Total cars in a parking lot (d) Number of siblings
116. **Which one is a property of joint probability distribution?**
- (a) $P(X_i, Y_j) < 1$ (b) $P(X_i, Y_j) = 0$ (c) $P(X_i, Y_j) < 0$ (d) $0 \leq P(X_i, Y_j) \leq 1$

X	0	1	2
P(x)	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

2.2 Situation Set

Answer the next two questions based on the following information

117. What is $F(1)$

- (a) 0.65 (b) 0.75 (c) 0.5 (d) 1

118. $P(X \leq 1 \leq 3) =$

- (a) 0.75 (b) 0.70 (c) 0.95 (d) 1

Answer the next three questions based on the following information

X	1	2	3
P(x)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

119. What is $P(X \geq 2)$?

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

120. Find the cumulative probability $F(2)$.

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

121. $P(X = 1 \text{ or } X = 3)$ equals

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

Answer the next three questions based on the following information

A fair six-sided die is rolled once. Let X be the number on the uppermost face.

122. What is $P(X > 4)$?

- (a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

123. Find $P(X \text{ is even})$.

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

124. $P(X \leq 2 \text{ or } X \geq 5)$ equals

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Answer the next three questions based on the following information

X	0	1	2	3
P(X)	$\frac{1}{4}$	m	$\frac{1}{3}$	$\frac{1}{6}$

125. What is the value of m?

- (a) $\frac{1}{3}$ (b) $\frac{5}{12}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$

126. Find $F(2)$.

- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

127. What is $P(X > 1)$?

- (a) $\frac{1}{2}$ (b) $\frac{5}{12}$ (c) $\frac{1}{3}$ (d) $\frac{7}{12}$

Answer the next three questions based on the following information

X	1	2	3	4	5
P(X)	$\frac{1}{5}$	c	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{3}$

128. What is the value of c?

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{20}$ (d) $\frac{1}{6}$

129. Find $P(2 < X \leq 4)$.

- (a) $\frac{5}{12}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

130. What is $P(X \leq 3)$?

- (a) $\frac{9}{20}$ (b) $\frac{7}{10}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

Answer the next three questions based on the following information

x	1	2	3
P(x)	$\frac{1}{3}$	a	$\frac{1}{6}$

131. What is the value of a?

- (a) $\frac{2}{3}$ (b) $\frac{5}{6}$ (c) $\frac{1}{2}$ (d) 1

132. Find $P(2 < X \leq 3)$

- (a) $\frac{5}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{6}$

133. What is $P(X \leq 3)$?

- (a) $\frac{5}{6}$ (b) $\frac{2}{5}$ (c) $\frac{1}{9}$ (d) $\frac{1}{7}$

Answer the next two questions based on the following information

x	1	2	3
P(x)	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

134. What is $F(2)$?

- (a) $\frac{2}{3}$ (b) $\frac{5}{6}$ (c) $\frac{1}{2}$ (d) 1

135. $P(1 < X \leq 2)$

- (a) $\frac{5}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{6}$

Answer the next two questions based on the following information

$$f(x) = kx; 0 < x < 5$$

136. What is the value of $P(2 < x < 3)$

- (a) $\frac{4}{5}$ (b) $\frac{3}{5}$ (c) $\frac{2}{5}$ (d) $\frac{1}{5}$

137. $P(X > 0)$

- (a) 0.99 (b) 0.5 (c) 1 (d) 0

Answer the next two questions using the following information

x	1	2	3	4	5	6
P(x)	k	2k	3k	4k	5k	6k

138. What is the value of k?

- (a) $\frac{7}{21}$ (b) $\frac{5}{21}$ (c) $\frac{1}{21}$ (d) 1

139. What is the type of variable X?

- (a) Discrete (b) Discrete random (c) Continuous (d) Continuous random

Answer the next THREE questions using the following information

$$P(x) = \frac{x+1}{k}; x = 1, 2, 3, 4$$

140. What is the value of k?

- (a) 10 (b) 11 (c) 14 (d) 15

141. $F(2) = -$

- (a) $\frac{2}{14}$ (b) $\frac{3}{11}$ (c) $\frac{5}{14}$ (d) $\frac{5}{11}$

142. $P(x)$ is a -

- (a) Joint probability distribution (b) Cumulative probability distribution
(c) Probability mass function (d) Probability Density function

143. $f(x) = 2x; 0 < X < 3$; What is $F(3)$?

- (a) 3 (b) 0 (c) 1 (d) 0

144. $f(x) = 3x; 0 < X < 2$; What is $F(2)$?

- (a) 6 (b) 3 (c) 1 (d) 0

145. $f(x) = x^2; 0 < X < 4$; What is $F(4)$?

- (a) 16 (b) 0 (c) 4 (d) 1

146. $f(x) = 4 - x; 1 < X < 5$; What is $F(5)$?

- (a) 3 (b) 0 (c) 1 (d) 4

Answer the next two questions based on the following information:

$$P(x, y) = \frac{1}{21}(x + y); x = 1, 2, 3 \text{ and } y = 1, 2$$

147. $P(x) = ?$

- (a) $P(x) = \frac{2x+3}{21}$ (b) $P(x) = \frac{x+3}{27}$ (c) $P(x) = \frac{4x+3}{21}$ (d) $P(x) = \frac{2x+5}{21}$

148. $P(y) = ?$

- (a) $\frac{y+2}{7}$ (b) $\frac{y+3}{7}$ (c) $\frac{3y+2}{7}$ (d) $\frac{y+2}{9}$

Answer the next two questions based on the following information.

x	4	5	6	3	2	1
P(X)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

149. The value of $P(3 < X < 5)$ is:

- (a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) 0

150. $P(x \neq 2)$ is :

- (a) $\frac{5}{6}$ (b) 0
(c) 1 (d) Can't be found from this information

2.3 Multiple Completion

151. If $f(x) = kx^3$; $-1 \leq x \leq 1$, then k is

- i) positive
- ii) negative
- iii) lies from -1 to 1

(a) i (b) ii (c) iii (d) i and ii

152. The example of a discrete random variable is–

- i. Binomial variate
- ii. Poisson variate
- iii. Normal variate

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

153. For a discrete random variable X with probability mass function $p(x)$:

- i. $P(a \leq X \leq b) = \sum_{x=a}^b p(x)$
- ii. $\sum_x p(x) = 1$
- iii. $p(x) \geq 0$ for all x

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

154. For any random variable X with cumulative distribution function $F(x)$:

- i. $F(x)$ is a non-decreasing function
- ii. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$
- iii. $P(X > a) = 1 - F(a)$

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

155. For the probability density function $f(x)$ of a continuous random variable X :

- i. $P(a \leq X \leq b) = \int_a^b f(x) dx$
- ii. $f(x) \geq 0$ for all x
- iii. $\int_{-\infty}^{\infty} f(x) dx = 1$

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

156. For a continuous random variable X with PDF $f(x) = 2x$, defined on $[0, 1]$:

- i. $f(x) \geq 0$ for all $x \in [0, 1]$
- ii. $\int_0^1 f(x) dx = 1$
- iii. $P(X > 1) = 0$

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

157. For a continuous random variable X with PDF $f(x) = k(2 - x)$ defined on $0 \leq x \leq 2$:

- i. The value of k is 1.
- ii. The cumulative distribution function $F(x) = x - \frac{x^2}{4}$ for $0 \leq x \leq 2$.
- iii. $P(1 < X < 2) = \frac{3}{8}$

Which one is correct?

- (a) i (b) i and ii (c) ii (d) i, ii and iii

3 Mathematical Expectation

158. $E(X+Y) = ?$
(a) $E(X) - E(Y)$ (b) $E(X) + E(Y)$ (c) $2E(X) - E(Y)$ (d) $E(X) \times E(Y)$
159. $E(4x+2Y) = ?$
(a) $E(X) - E(Y)$ (b) $4E(X) + 2E(Y)$ (c) $2E(X) + 4E(Y)$ (d) $E(X) \times E(Y)$
160. If $V(X) = 2$, what is the variance of $(X + 2m)$, where m is a constant?
(a) 0 (b) 2 (c) 4 (d) $2 + 4m$
161. If $V(X) = 2$, what is the variance of $(2X + m)$, where m is a constant?
(a) 0 (b) 2 (c) 4 (d) 8
162. What is the expected value of the squared deviation of the value of the random variable from their mean?
(a) Arithmetic Mean (b) Expectation (c) Variance (d) Co-variance
163. What is the minimum value of variance a random variable?
(a) $-\infty$ (b) 1 (c) 0 (d) -1
164. If $y = ax + b$, what is the value of $V(y)$?
(a) $aV(X)$ (b) $a^2V(X)$ (c) $V(X)$ (d) a^2
165. If $y = ax + b$, what is the value of $E(y)$?
(a) $aE(X) + b$ (b) $a^2E(X)$ (c) $E(X)$ (d) b
166. What is the value of $V(5)$?
(a) 0 (b) 25 (c) 5 (d) 1
167. If $P(x) = \frac{1}{n}; x = 1, 2, 3, \dots, n$, what is the value of $E(X)$?
(a) $\frac{n}{2}$ (b) $\frac{n-1}{2}$ (c) $\frac{n+1}{2}$ (d) $n + 1$
168. If $P(x) = \frac{4 - |5 - x|}{k}; x = 2, 3, 4, \dots, 8$, what is the value of k ?
(a) 5 (b) 8 (c) 16 (d) 24
169. Expected value of a constant a is –
(a) 1 (b) Variance (c) a (d) $a+1$
170. The variance of a constant m is –
(a) 0 (b) 1 (c) m (d) m^2
171. What is $V(X - Y)$ equal to?
(a) $V(X) + V(Y)$ (b) $V(X) + V(Y) - 2Cov(X, Y)$
(c) $V(X) - V(Y)$ (d) $V(X) + V(Y) + 2Cov(X, Y)$
172. What is the value of $V(2X+5)$?
(a) $4V(X) - 5$ (b) 20 (c) $4V(X)$ (d) 0
173. If $P(x) = \frac{1}{20}; x = 1, 2, 3, \dots, 20$, what is the standard deviation?
(a) 1 (b) 5.77 (c) 7.75 (d) 12.57

174. **Expectation measures –**

- (a) Dispersion (b) Skewness (c) Kurtosis (d) Central tendency

175. **If $E(X) = -0.5$, then $E(1 - 2X) = ?$**

- (a) 0 (b) -1 (c) 2 (d) 1

176. **If $P(X) = \frac{1}{10}; x = 1, 2, \dots, 10$, then $E(X) = ?$**

- (a) 10 (b) 5.5 (c) 0 (d) 11

177. **Which formula of variance is correct?**

- (a) $V(X + Y) = V(X) + V(Y) - 2Cov(X, Y)$ (b) $V(X + Y) = V(X) + V(Y) + 2Cov(X, Y)$
(c) $V(X + Y) = V(X) + V(Y) - 2Cov(X, Y)$ (d) $V(X + Y) = V(X) - V(Y) + 2Cov(X, Y)$

178. **X is a constant; what is the value of $V(\frac{X}{2})$?**

- i) 0
ii) $\frac{1}{2}$
iii) $\frac{1}{4}$

- (a) ii (b) i (c) iii (d) i and iii

179. **If $E(X) = 2, E(X^2) = 8, V(X) = --$**

- (a) 0 (b) 2 (c) 4 (d) 8

180. **If $E(X) = 3, E(X^2) = 11, V(X) = --$**

- (a) 2 (b) 5 (c) 6 (d) 4

181. **If $E(X) = 4$, what is $E(3X - 2)$?**

- (a) 10 (b) 8 (c) 7 (d) 6

182. **If $E(X) = 5, E(X^2) = 30, V(X) = --$**

- (a) 3 (b) 5 (c) 4 (d) 6

183. **If $E(X) = 6$, what is $E(\frac{X}{2} + 1)$?**

- (a) 4 (b) 3 (c) 2 (d) 5

184. **If $E(X) = 2, E(X^2) = 10, V(X) = --$**

- (a) 5 (b) 6 (c) 7 (d) 4

185. **If $E(X) = 7$, what is $E(4X + 3)$?**

- (a) 28 (b) 30 (c) 31 (d) 29

186. **If $E(X) = 3$, what is $E(5 - X)$?**

- (a) 2 (b) 3 (c) 4 (d) 5

187. **If $E(X) = 4$ and $V(X) = 5$, what is $E(X^2)$?**

- (a) 9 (b) 16 (c) 21 (d) 25

188. **If $E(X) = 3$ and $V(X) = 7$, what is $E(X^2)$?**

- (a) 9 (b) 10 (c) 16 (d) 18

189. **If $E(X) = 5$ and $E(X^2) = 34$, what is $V(X)$?**

- (a) 6 (b) 9 (c) 10 (d) 7

190. If $E(X) = 2$ and $E(X^2) = 14$, what is $V(X)$?
 (a) 10 (b) 9 (c) 8 (d) 7
191. If $E(X) = 6$ and $V(X) = 12$, what is $E(X^2)$?
 (a) 36 (b) 40 (c) 48 (d) 50
192. If $P(x) = \frac{4-|5-x|}{k}; x = 2, 3, 4, \dots, 8$, what is the value of $E(X)$?
 (a) 3 (b) 8 (c) 16 (d) 5
193. If $P(x) = \frac{6-|7-x|}{k}; x = 2, 3, 4, \dots, 12$, what is the value of $E(X)$?
 (a) 6 (b) 9 (c) 13 (d) 36
194. If $P(x) = \frac{3-|4-x|}{k}; x = 2, 3, 4, \dots, 6$, what is the value of k ?
 (a) 6 (b) 9 (c) 10 (d) 40
195. If the variance of X is 3, what is the variance of $V(3)$?
 (a) 1 (b) 2 (c) 3 (d) 0
196. If $V(X) = 5$, what is $V(X + 5)$?
 (a) 0 (b) 5 (c) 10 (d) 25
197. If $V(X) = 5$, what is $V(2X + 5)$?
 (a) 20 (b) 5 (c) 10 (d) 25
198. If $E(X) = 2$ and $E(X^2) = 8$, then the value of the $V(X) = ?$
 (a) 0 (b) 2 (c) 4 (d) 8
199. If $E(X^2) = 20$ and $V(X) = 11$, what is $E(X)$?
 (a) 3 (b) 4 (c) 5 (d) 6
200. If $E(X^2) = 50$ and $V(X) = 14$, what is $E(X)$?
 (a) 5 (b) 6 (c) 7 (d) 8
201. If $E(X^2) = 25$ and $V(X) = 9$, what is $E(X)$?
 (a) 2 (b) 3 (c) 4 (d) 5
202. If $E(X^2) = 45$ and $V(X) = 21$, what is $E(X)$?
 (a) $4\sqrt{3}$ (b) $2\sqrt{6}$ (c) $6\sqrt{2}$ (d) $7\sqrt{2}$
203. If $E(X^2) = 13$ and $V(X) = 4$, what is $E(X)$?
 (a) 2 (b) 3 (c) 4 (d) 5
204. If $E(X) = 3$, what is $E(2X - 5)$?
 (a) 1 (b) 3 (c) 5 (d) 7
205. If $E(X) = 4$, what is $E(\frac{X}{2} + 3)$?
 (a) 4 (b) 5 (c) 6 (d) 7
206. If $E(X) = -2$, what is $E(3X + 7)$?
 (a) 1 (b) -1 (c) -2 (d) 4
207. If $E(X) = 6$, what is $E(5 - X)$?
 (a) 1 (b) 0 (c) -1 (d) 2

208. If $E(X) = 10$, what is $E(4X - 8)$?
- (a) 12 (b) 40 (c) 28 (d) 32
209. If $P(x) = \frac{1}{15}; x = 1, 2, 3, \dots, 15$, what is the value of the expectation?
- (a) 8.5 (b) 7.5 (c) 7 (d) 8

3.1 Situation Set

Answer the next **THREE** questions based on the following information

X	0	1	2
P(x)	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{5}{12}$

210. What is the value of $E(X)$?
- (a) $\frac{15}{12}$ (b) $\frac{13}{12}$ (c) $\frac{1}{12}$ (d) $\frac{11}{13}$
211. What is the value of $E(X^2)$?
- (a) $\frac{25}{12}$ (b) $\frac{13}{12}$ (c) $\frac{23}{12}$ (d) $\frac{25}{13}$
212. What is $V(2X)$?
- (a) 2.93 (b) 2.91 (c) 1.97 (d) 2.97

Answer the next **TWO** questions based on the following information

X	1	2	4
P(x)	0.5	0.3	0.2

213. What is $E(X)$?
- (a) 1.5 (b) 1.9 (c) 2.0 (d) 2.1
214. What is $V(X)$?
- (a) 1.09 (b) 1.29 (c) 1.49 (d) 1.69

Answer the next **TWO** questions based on the following information

X	1	2	3
P(x)	0.5	0.3	0.2

215. What is $E(X^2)$?
- (a) 3.6 (b) 4.2 (c) 5.0 (d) 5.6
216. What is $V(2X)$?
- (a) 2.24 (b) 3.36 (c) 4.48 (d) 5.60

Answer the next **TWO** questions based on the following information

217. What is $E(X)$?
- (a) 2.5 (b) 3.4 (c) 4.0 (d) 4.6
218. What is $V(X)$?
- (a) 2.04 (b) 2.44 (c) 2.84 (d) 3.24

Answer the next **TWO** questions based on the following information

X	2	4	6
P(x)	0.5	0.3	0.2

X	0	1	4
P(x)	0.4	0.4	0.2

219. What is $E(X^2)$?

- (a) 1.2 (b) 2.0 (c) 3.6 (d) 4.8

220. What is $V(X)$?

- (a) 2.24 (b) 2.64 (c) 3.04 (d) 3.44

Answer the next TWO questions based on the following information

The probability mass function of a discrete random variable X is given by

$$P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

221. What is $E(X)$?

- (a) 2.0 (b) 2.5 (c) 3.0 (d) 3.5

222. What is $V(X)$?

- (a) 0.50 (b) 0.75 (c) 1.00 (d) 1.25

Answer the next TWO questions based on the following information

The probability mass function of a discrete random variable X is given by

$$P(X = x) = \frac{x}{6}, \quad x = 1, 2, 3.$$

223. What is $E(X)$?

- (a) 1.33 (b) 2.00 (c) 2.33 (d) 3.00

224. What is $V(X)$?

- (a) 0.33 (b) 0.56 (c) 0.89 (d) 1.11

Answer the next THREE questions based on the following information

X	1	2	3
P(x)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

225. What is the value of $E(X)$?

- (a) 2.00 (b) 2.17 (c) 2.33 (d) 2.50

226. What is the value of $E(X^2)$?

- (a) 5.17 (b) 4.83 (c) 5.00 (d) 5.33

227. What is $V(3X)$?

- (a) 9.67 (b) 11.33 (c) 12.67 (d) 4.25

Answer the next two questions based on the following information

The probability function of random variable x is given below:

$$P(x) = \frac{x}{k}; x = 1, 2, 3, 4$$

228. What is the value of k ?

- (a) 6 (b) 10 (c) 15 (d) 20

229. What is $E(X)$?

- (a) 2.25 (b) 3.5 (c) 2.5 (d) 3.0

Answer the next three questions based on the following information

The probability function of random variable x is given below:

$$P(x) = \frac{2x+1}{k}; x = 1, 2, 3, 4$$

230. What is the value of k ?

- (a) 18 (b) 25 (c) 12 (d) 24

231. What is $E(X)$?

- (a) 1.75 (b) 2.92 (c) 3.25 (d) 2.25

232. What is $V(X)$?

- (a) 1.05 (b) 3.0 (c) 1.5 (d) 1.25

Answer the next two questions based on the following information

The probability function of random variable x is given below

$$P(x) = \frac{x-1}{k}; x = 2, 3, 4, 5$$

233. What is the value of k ?

- (a) 2 (b) 5 (c) 10 (d) 25

234. What is $E(X)$?

- (a) 0.425 (b) 0.525 (c) 0.725 (d) 0.625

3.2 Multiple Completion

235. The possible relationship between $E(X)$ and $E(X^2)$

- i. $E(X) \geq E(X^2)$
ii. $E(X) \leq E(X^2)$
iii. $E(X) = E(X^2)$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

4 Binomial Distribution

236. How many parameters are there in a binomial distribution?

- (a) 1 (b) 2 (c) 3 (d) 4

237. In a Binomial distribution, how are mean and variance related?

- (a) $Mean > Variance$ (b) $Mean < Variance$
(c) $Mean = Variance$ (d) $Mean = 2 \times Variance$

238. What is the Mean of Binomial Distribution?
 (a) np (b) npq (c) nq (d) $\sqrt{npq}$
239. What is the Variance of Binomial Distribution?
 (a) np (b) npq (c) nq (d) $\sqrt{npq}$
240. What is the 2nd central moment of binomial distribution?
 (a) np (b) npq (c) $\sqrt{npq}$ (d) p
241. What is the Standard Deviation of Binomial Distribution?
 (a) np (b) npq (c) nq (d) $\sqrt{npq}$
242. What is the coefficient of variation of binomial distribution?
 (a) np (b) npq (c) $\sqrt{npq}$ (d) $\sqrt{\frac{q}{np}}$
243. Which is true of mean (np) of Binomial Distribution?
 (a) $np = 0$ (b) $np < 0$ (c) $np > 0$ (d) $np \neq 0$
244. Which of the following is correct in the case of binomial distribution?
 (a) Number of parameter is one (b) Mean is greater than variance
 (c) Number of trials is more than 30 (d) Always negative skewness
245. What is true of binomial distribution?
 (a) There is one parameter (b) Number of trial is fixed
 (c) Mean is greater than variance (d) Skewness is negative
246. How many possible outcome in binomial experiment?
 (a) 2 (b) 3 (c) 4 (d) 30
247. Which of the following is recursion formula of binomial distribution?
 (a) $P(x+1) = \frac{n-x}{x+1} \frac{p}{q} P(x); x = 0, 1, 2, \dots, n$ (b) $P(x+1) = \frac{n+x}{x+1} \frac{p}{q} P(x); x = 0, 1, 2, \dots, n$
 (c) $P(x-1) = \frac{n-x}{x+1} \frac{p}{q} P(x); x = 0, 1, 2, \dots, n$ (d) $P(x) = \frac{n-x}{x+1} \frac{p}{q} P(x+1); x = 0, 1, 2, \dots, n$
248. Which of the following is the skewness of binomial distribution?
 (a) $\beta_1 = \frac{(q-p)^2}{npq}$ (b) $\beta_1 = \frac{(q-p)^2}{np}$ (c) $\beta_1 = \frac{(q-p)^2}{nq}$ (d) $\beta_1 = \frac{(n-p)^2}{npq}$
249. Which of the following is the kurtosis of binomial distribution?
 (a) $\beta_2 = 3 + \frac{1-6pq}{npq}$ (b) $\beta_2 = 3 + \frac{1-6nq}{npq}$ (c) $\beta_2 = 3$ (d) $\beta_2 = 3 - \frac{1-npq}{npq}$
250. When binomial distribution is symmetric distribution?
 (a) $p \neq q$ (b) $p \neq q$ (c) $p \leq q$ (d) $p = q$
251. When binomial distribution is tends to poisson distribution?
 (a) $n \rightarrow \infty$ and $p \rightarrow 0$ (b) $n \rightarrow 0$ and $p \rightarrow 0$ (c) $n \rightarrow \infty$ and $p \rightarrow \infty$ (d) $n \rightarrow \infty$ and $p = 0$
252. When binomial distribution is tends to normal distribution?
 (a) $n \rightarrow \infty$ and $p \rightarrow 0$ (b) $n \rightarrow 0$ and $p \rightarrow 0$ (c) $n \rightarrow \infty$ and $p \rightarrow \infty$ (d) $n \rightarrow \infty$ and $p \cong q$
253. When binomial distribution is positive skewness?
 (a) $p \neq q$ (b) $p \neq q$ (c) $p \leq q$ (d) $p = q$

254. When binomial distribution is negative skewness?
 (a) $p < q$ (b) $p > q$ (c) $p \leq q$ (d) $p = q$
255. Which of the following is the number of obtained heads x in the 12 times coin experiment?
 (a) Binomial variate (b) Poisson variate (c) Normal variate (d) Bivariate
256. In a binomial distribution with $p = 0.3$ and $n = 10$, what is $P(2)$?
 (a) 0.2335 (b) 0.2668 (c) 0.3828 (d) 0.1211
257. In a binomial distribution with $p = 0.4$ and $n = 12$, what is $P(3)$?
 (a) 0.0896 (b) 0.2131 (c) 0.1419 (d) 0.2942
258. In a binomial distribution with $p = 0.5$ and $n = 8$, what is $P(4)$?
 (a) 0.2734 (b) 0.3125 (c) 0.2070 (d) 0.0898
259. In a binomial distribution with $p = 0.2$ and $n = 15$, what is $P(5)$?
 (a) 0.1789 (b) 0.1887 (c) 0.1032 (d) 0.2413
260. In a binomial distribution with $p = 0.6$ and $n = 9$, what is $P(6)$?
 (a) 0.2007 (b) 0.2508 (c) 0.2311 (d) 0.7682
261. In a binomial distribution with $p = 0.3$ and $P(x) = 0.2508, n = 9, x = ?$
 (a) 18 (b) 10 (c) 13 (d) 6
262. In a binomial distribution with $p = 0.4$ and $P(x) = 0.1419$, what is n ?
 (a) 5 (b) 6 (c) 12 (d) 15
263. In a binomial distribution with $p = 0.5$ and $P(2) = 0.1093$, what is n ?
 (a) 15 (b) 1 (c) 8 (d) 12
264. In a binomial distribution with $p = 0.2$ and $P(x) = 0.9389, n = ?$
 (a) 7 (b) 12 (c) 11 (d) 15
265. In a binomial distribution with $p = 0.6$ and $P(5) = 0.02449, n = ?$
 (a) 3 (b) 9 (c) 10 (d) 15

4.1 Situation Set

Answer the next two questions based on the following information.

X is a binomial variate with expectation 4 and standard deviation $\sqrt{3}$.

266. What are the values of the parameters (mean and probability)?
 (a) $16, \frac{1}{4}$ (b) $16, \frac{3}{4}$ (c) $15, \frac{1}{4}$ (d) $10, \frac{1}{4}$
267. What is $P(X \neq 0)$?
 (a) 0 (b) 0.01 (c) 0.99 (d) 1

Answer the next three questions based on the following information

In a binomial distribution, $P(x = 4) = \frac{1}{2}P(x = 5)$ and $n = 10$.

268. What is the mean of the random variable x ?
 (a) 0.75 (b) 1.50 (c) 3.75 (d) 6.25

269. $P(x = 2) = \text{---}$

- (a) 0.0053 (b) 0.0069 (c) 0.0085 (d) 0.94

270. **The mathematical form of the distribution of the random variable x is -**

- i. $P(x) = {}^{10}C_x (0.625)^x (0.375)^{10-x}, x = 0, 1, 2, \text{---}, 10$
ii. $P(x) = {}^{10}C_x \left(\frac{5}{8}\right)^x \left(\frac{3}{8}\right)^{10-x}; x = 0, 1, 2, \text{---}, 10$
iii. $P(x) = {}^{10}C_x (0.625)^x (0.375)^{x-10}; x = 0, 1, 2, \text{---}, \infty$

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

Answer the next three questions based on the following information

The values of the parameters n and p of a binomial variate (x) are obtained 5 and 0.2 respectively.

271. **What is the value of the 2nd central moment of the binomial variate?**

- (a) 0.20 (b) 0.79 (c) 0.80 (d) 1.00

272. **What is the probability of non zero of the binomial variate?**

- (a) 0.5723 (b) 0.6723 (c) 0.8823 (d) 1.00

273. **The nature and shape of the binomial variate(x) is -**

- (a) Positive skewness and leptokurtic (b) Positive skewness and platykurtic
(c) Positive skewness and mesokurtic (d) Negative skewness and leptokurtic

Answer the next two questions based on the following information

If an unbiased coin is tossed 6 times, then the random variable x is denote the number of the obtained heads on the top face.

274. **What is the coefficient of skewness of the random variable x is -**

- (a) 0 (b) 1.50 (c) 2.75 (d) 3.00

275. **The distribution of the random variable x is -**

- i. Symmetrical
ii. Platykurtic
iii. Leptokurtic

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

Answer the next two questions based on the following information

About 30% items of the produced bulbs are defective in a factory. Among the 1000 packets, 10 bulbs are packed in one packet.

276. **How many packets whose contains no defective bulbs?**

- (a) 28 (b) 30 (c) 75 (d) 100

277. **The probability of the defective bulbs follows that distribution whose nature and shape is -**

- (a) Positive skewness and leptokurtic (b) Positive skewness and platykurtic
(c) Positive skewness and mesokurtic (d) Negative skewness and leptokurtic

Answer the next two questions based on the following information

Bangladesh and India play three test matches. The probability that Bangladesh wins a cricket test match against India is given to be $\frac{1}{3}$.

278. What is the probability that Bangladesh will win at least one test match?

- (a) 0.2036 (b) 0.3020 (c) 0.5075 (d) 0.7037

279. The probability of Bangladesh will win test match follows that distribution whose nature and shape

- (a) Positive skewness and leptokurtic (b) Positive skewness and platykurtic
(c) Positive skewness and mesokurtic (d) Negative skewness and leptokurtic

Answer the next THREE questions based on the following information

The mean of a Binomial distribution is 40 and standard deviation 6.

280. What is the value of n ?

- (a) 200 (b) 300 (c) 400 (d) 500

281. What is the value of $1 - q$?

- (a) 0.5 (b) 0.2 (c) 0.3 (d) 0.1

282. What is the value of $P(X \leq 40)$?

- (a) 0.52 (b) 0.54 (c) 0.45 (d) 0.91

Answer the next questions based on the following information

35% examinee succeed in the admission test examination of 'Ka' unit and 15% examinee succeed in the admission test examination of 'Kha' unit of a university in 2015. HSC passed 25 students of 'Ka' unit and 20 students of 'Kha' unit are participants in the admission test examination from a college in that year.

283. What is the probability of succeed 10 students of 'Ka' unit?

- (a) 0.035 (b) 0.115 (c) 0.140 (d) 0.75

284. What is the probability of succeed 5 students of 'Kha' unit?

- (a) 0.035 (b) 0.103 (c) 0.115 (d) 0.140

285. The probability of succeed students of 'Ka' unit follows that distribution whose mean is-

- (a) 8.75 (b) 15 (c) 20 (d) 25

286. The probability of succeed students of 'Kha' unit follows that distribution whose mean is-

- (a) 3 (b) 15 (c) 20 (d) 25

287. The probability of succeed students of 'Ka' unit follows that distribution whose variance is-

- (a) 2.25 (b) 5.69 (c) 8.75 (d) 25

288. The probability of succeed students of 'Kha' unit follows that distribution whose variance is-

- (a) 2.25 (b) 3 (c) 8.75 (d) 25

289. The probability of succeed students of 'Ka' unit follows that distribution whose SD is-

- (a) 2.38 (b) 5.69 (c) 8.75 (d) 25

290. The probability of succeed students of 'Kha' unit follows that distribution whose SD is-
 (a) 1.60 (b) 3.00 (c) 8.75 (d) 25.00
291. The probability of succeed students of 'Ka' unit follows that distribution whose skewness is-
 (a) 0.02 (b) 2.38 (c) 5.69 (d) 8.75
292. The probability of succeed students of 'Kha' unit follows that distribution whose skewness is-
 (a) 0.19 (b) 1.60 (c) 3.00 (d) 8.75
293. The probability of succeed students of 'Ka' unit follows that distribution whose kurtosis is-
 (a) 0.02 (b) 2.38 (c) 2.94 (d) 8.75
294. The probability of succeed students of 'Kha' unit follows that distribution whose kurtosis is-
 (a) 0.19 (b) 1.60 (c) 3.09 (d) 8.75
295. The probability of succeed students of 'Ka' unit follows that distribution whose distribution is-
 (a) mean $\neq$ variance (b) mean = variance (c) mean = 0 (d) mean $\neq$ variance
296. The probability of succeed students of 'Kha' unit follows that distribution whose distribution is-
 (a) mean $\neq$ variance (b) mean = variance (c) mean = 0 (d) mean $\neq$ variance
297. The probability of succeed students of 'Ka' unit follows that distribution whose nature and shape is-
 (a) Positive skewness and leptokurtic (b) Positive skewness and platykurtic
 (c) Positive skewness and mesokurtic (d) Negative skewness and leptokurtic
298. The probability of succeed students of 'Kha' unit follows that distribution whose nature and shape is-
 (a) Positive skewness and leptokurtic (b) Positive skewness and platykurtic
 (c) Positive skewness and mesokurtic (d) Negative skewness and leptokurtic

4.2 Multiple Completion

299. The characteristics of binomial distribution—

- i. $E(X) > V(X)$
- ii. $E(X) = V(X)$
- iii. $E(X) = np$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

300. In a binomial distribution with parameters n and p :

- i. The expected value is given by $E(X) = np$.
- ii. The variance is given by $V(X) = np(1 - p)$.
- iii. The standard deviation is given by $\sqrt{np}$.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

301. Which of the following statements about a binomial distribution are true?

- i. The probability of success remains constant for each trial.
- ii. The trials are dependent on each other.
- iii. The number of trials is fixed in advance.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

302. Consider a binomial experiment. Which of the following statements is/are true?

- i. Each trial results in exactly one of two possible outcomes.
- ii. The expected value is always greater than the variance.
- iii. The probability mass function of a binomial distribution can be computed using the binomial formula.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

303. Which of the following is/are correct about the binomial distribution?

- i. The variance is maximized when $p = 0.5$.
- ii. If $p = 1$, the distribution becomes degenerate.
- iii. The standard deviation is given by $\sqrt{np(1-p)}$.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

304. If an unbiased coin is tossed at once, then it is -

- i. Bernoulli's trial
- ii. Binomial trial
- iii. Unit trial

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

305. The internal conditions of binomial distribution are -

- i. Number of trials is finite
- ii. Number of trials is more than 100
- iii. The trials are independent of each other

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

306. Binomial distribution is used to -

- i. test ratio of large sample
- ii. test ratio of small sample
- iii. fit the frequency distribution

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

307. The trials of binomial experiments are -

- i. independent of each other
- ii. dependent of each other
- iii. not dependent of each other

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

308. The mathematical form of binomial distribution are -

- i. $P(x) = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$
- ii. $P(x) = {}^mC_x p^x q^{m-x}; x = 0, 1, 2, \dots, m$
- iii. $P(x) = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, \infty$

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

309. The two parameters of binomial distribution are -

- i. n and p
- ii. n and q
- iii. p and q

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

310. The probability of successes and failures at each trial in binomial distribution are -

- i. same
- ii. constant
- iii. different

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

311. The kurtosis of binomial distribution are -

- i. Leptokurtic
- ii. Mesokurtic
- iii. Platykurtic

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

312. The mode of binomial distribution may be -

- i. one
- ii. two
- iii. three

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

313. In the case of binomial distribution -

- i. The values of n, p and q are non negative
- ii. The values of n, p and q are positive
- iii. $n \geq 1, p \geq 0$ and $q \geq 0$

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

314. The mean of binomial distribution may be -

- i. non negative
- ii. Positive
- iii. More than zero

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

315. The binomial distribution is -

- i. Symmetrical
- ii. Positive Skewness
- iii. Negative Skewness

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

316. The uses of binomial distribution are -

- i. Cancer research
- ii. Determination of the defective product
- iii. Analysis of time series

Which of the following is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

5 Poisson Distribution

317. The no. of parameters in a Poisson distribution is —

- (a) 1 (b) 2 (c) 3 (d) 4

318. What is the mean of Poisson distribution

- (a) $\frac{1}{\sqrt{m}}$ (b) m (c) $\frac{1}{m}$ (d) $1 + \frac{1}{m}$

319. Which relationship between mean and variance of Poisson Distribution is correct?

- (a) $Mean > Variance$ (b) $Mean < Variance$ (c) $Mean = Variance$ (d) $Mean \neq Variance$

320. What is the Variance of Poisson Distribution(with parameter m)?

- (a) $\frac{1}{\sqrt{m}}$ (b) $\frac{1}{m}$ (c) m (d) $\frac{1}{m+1}$

321. What is the Standard Deviation of Poisson Distribution(with parameter m)?

- (a) $\frac{1}{\sqrt{m}}$ (b) $\frac{1}{m}$ (c) $\sqrt{m}$ (d) $\frac{1}{m+1}$

322. Which one is true of the parameter (m) of Poisson Distribution?

- (a) $m = 0$ (b) $m < 0$ (c) $m > 0$ (d) $m = 1$

323. The parameter of a Poisson Distribution is 5. What is its mean?

- (a) 2 (b) 5 (c) 2.24 (d) 25

324. When does Binomial Distribution tend to Poisson Distribution?

- (a) $n \rightarrow \infty, p \rightarrow 0$ & np is finite (b) $n \rightarrow \infty, p \rightarrow 0$ & np is infinite
(c) $n \rightarrow \infty, p \rightarrow \infty$ & np is finite (d) $n \rightarrow 0, p \rightarrow \infty$ & np is infinite

325. The parameter of a Poisson variate is 2. What is its variance?

- (a) 0 (b) 4 (c) $\sqrt{2}$ (d) 2

326. The parameter of a Poisson variate is 5. What is its variance?

- (a) 10 (b) 5 (c) $\sqrt{5}$ (d) 25

327. A Poisson distribution has a mean of 3. What is the variance?

- (a) 9 (b) 3 (c) $\sqrt{3}$ (d) 0

328. **X is a Poisson variate. $P(2) = P(4)$. What is the value of the parameter?**
 (a) 12 (b) 3.46 (c) 3.6 (d) 4
329. **X is a Poisson variate. $P(3) = P(5)$. What is the value of the parameter?**
 (a) 4.5 (b) 5 (c) 2.3 (d) 4.1
330. **For a Poisson variate X, if $P(1) = P(3)$, what is the variance?**
 (a) 2.5 (b) 3.2 (c) 2.45 (d) 4.5
331. **For a Poisson variate X, if $P(2) = P(3)$, what is the variance?**
 (a) 3 (b) 4 (c) 5 (d) 6
- Answer the next two questions based on the following information**
 For a Poisson variate X, if $P(2) = P(5)$.
332. **What is standard deviation?**
 (a) 1.978 (b) 1.998 (c) 1.989 (d) 1.889
333. **What is the value of $P(2)$?**
 (a) 0.25 (b) 0.14 (c) 0.15 (d) 0.02
334. **The standard deviation of a poisson distribution is 2. What is the parameter?**
 (a) 2 (b) 3 (c) 4 (d) 5
335. **Mean of a Poisson variate is a. What is its standard deviation?**
 (a) 0 (b) a (c) $a^{\frac{1}{2}}$ (d) a^2
336. **The standard deviation of a Poisson distribution is 3. What is the parameter?**
 (a) 6 (b) 9 (c) 3 (d) 4
337. **For a Poisson distribution with a mean of 5, what is the variance?**
 (a) 5 (b) 10 (c) 25 (d) 15
338. **If the variance of a Poisson distribution is 4, what is $P(2)$?**
 (a) 0.1465 (b) 0.1954 (c) 0.1839 (d) 0.2184
339. **If the variance of a Poisson distribution is 3.5, what is $P(1)$?**
 (a) 0.1465 (b) 0.1057 (c) 0.1839 (d) 0.2184
340. **A Poisson distribution has a mean of 7. What is the standard deviation?**
 (a) 3.2 (b) 4.1 (c) 2.65 (d) 1.78
341. **If $P(2)$ in a Poisson distribution with parameter λ equals 0.2240, what is the parameter λ ?**
 (a) 2.4551 (b) 1.2515 (c) 1.2115 (d) 2.5112
342. **A Poisson distribution has a mean of 4. What is $P(3)$?**
 (a) 0.1465 (b) 0.1954 (c) 0.1839 (d) 0.2381
343. **If the variance of a Poisson distribution is 3, what is the mean?**
 (a) 3 (b) $\sqrt{3}$ (c) 2 (d) 6
344. **For a Poisson distribution with mean 6, what is the probability of $P(0)$?**
 (a) 0.0895 (b) 0.012 (c) 0.0454 (d) 0.0024

345. The mean of a Poisson distribution is 10. What is its standard deviation?
 (a) 5 (b) $\sqrt{10}$ (c) 10 (d) $\sqrt{20}$
346. Given that the parameter of a Poisson distribution is 8, what is the variance?
 (a) 4 (b) 8 (c) $\sqrt{8}$ (d) 16

5.1 Multiple Completion

347. For a Poisson-distributed variable with mean $\lambda = 4$, which of the following is true?
 i. $E(X) = 4$
 ii. $V(X) = 2$
 iii. $E(X^2) = 18$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
348. If $X \sim \text{Poisson}(m = 3)$, which of the following holds?
 i. $E(X) = 3$
 ii. $V(X) = 3$
 iii. $E(X^2) = 12$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
349. For a Poisson distribution, which of the following statements are true?
 i. The mean and variance are always equal.
 ii. The distribution is always symmetric.
 iii. The probability of zero occurrences is given by e^{-m} .

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
350. A Poisson-distributed random variable has mean $\lambda = 6$. Among the following properties —
 i. $E(X) = 6$
 ii. $V(X) = 6$
 iii. $P(X = 0) = e^{-6}$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
351. For a Poisson process with $\lambda = 5$, which of the following is true?
 i. The standard deviation is $\sqrt{5}$.
 ii. $P(X \geq 1) = 1 - e^{-5}$.
 iii. $E(X^2) = 30$

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
- ## 5.2 Problems
352. On average, 1 in 1000 houses in a city gets a fire-burn in a year. If there are 2000 houses, what is the probability that, in a certain year, exactly 5 house will be burnt?
 (a) 0.036 (b) 0.040 (c) 0.027 (d) 0.091

6 Normal Distribution

353. What is the range of the Normal distribution?
(a) -1 to 1 (b) 0 to 1 (c) 0 to ∞ (d) $-\infty$ to ∞
354. What is the probability of a certain value of the Normal distribution?
(a) 0 (b) 1 (c) 0.5 (d) 0.1
355. What is the range of the mean of the Normal distribution?
(a) $(-\infty, \infty)$ (b) $(0, \infty)$ (c) $(0, 1)$ (d) $(-\infty, 0)$
356. What is the relationship between the normal and standard normal distribution?
(a) $Z = \frac{X+\mu}{\sigma}$ (b) $Z = \frac{X-\mu}{\sigma}$ (c) $Z = \frac{X-\sigma}{\mu}$ (d) $Z = \frac{X-\mu}{\sigma^2}$
357. What is the mean of standard normal distribution?
(a) -1 (b) 1 (c) 0 (d) ∞
358. In the standard normal distribution, how many parameters are there?
(a) 1 (b) 2 (c) 0 (d) 3
359. What is the maximum value of the probability density function of a normal distribution?
(a) μ (b) $\frac{1}{\sigma\sqrt{2\pi}}$ (c) σ (d) σ^2
360. What is the skewness of the Normal distribution?
(a) 0 (b) 1 (c) 2 (d) 3
361. How many parameters does the Normal distribution have?
(a) 0 (b) 1 (c) 2 (d) 3

6.1 Mathematical Problems

Answer the next two questions based on the following information

A Normal distribution has mean 12 and standard deviation 2 .

362. $P(10 < X < 13.5) =$ —
(a) 0.50 (b) 0.56 (c) 0.64 (d) 0.45
363. $P(X \text{ is between } 6 \text{ and } 18) =$ —
(a) 0.002 (b) 0.976 (c) 0.997 (d) 0.590

7 Index Number

364. If Laspeyres' index is 80 and Fisher's index is 120 , what is the value of Paasche's index?
(a) 97.97 (b) 100 (c) 180 (d) 200
365. The year for which an index number is calculated is called —
(a) Base year (b) Index year (c) Quantity year (d) Current year

366. Which Index Number satisfies the time and factor reversal test?
- (a) Laspeyres' index number (b) Paasche's index number
(c) Fisher's index number (d) Marshal-Edgeworth index number

367. Which index number shows an upward trend?
- (a) Laspeyres' index number (b) Bowley's index number
(c) Fisher's index number (d) Marshal-Edgeworth index number

8 Sampling

368. A sample characteristic is known as a –
- (a) Quantity (b) Statistic (c) Parameter (d) Unit
369. The complete enumeration of all items in a target population is –
- (a) Pilot survey (b) Sample survey (c) Census (d) Sampling frame
370. An unknown constant of the population is called –
- (a) Quantity (b) Statistic (c) Parameter (d) Unit
371. In which sampling method are all items equally likely
- (a) Simple random sampling (b) Quota sampling
(c) Stratified sampling (d) Cluster sampling
372. How many samples of size 4 can be drawn with replacement from a population of size 10?
- (a) 200 (b) 220 (c) 210 (d) 10000
373. How many samples of size 4 can be drawn without replacement from a population of size 10?
- (a) 200 (b) 220 (c) 210 (d) 550
374. In sampling with replacement, the probability of each selected sample is –
- (a) $\frac{1}{N^n}$ (b) $\frac{1}{N C_n}$ (c) $\frac{1}{N \times n}$ (d) $\frac{n}{N}$
375. Which of the following are true about stratified sampling?
- i. The population is divided into homogeneous subgroups called strata
ii. Samples are drawn independently from each stratum
iii. The strata are heterogeneous
- Which one is correct?
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

9 Vital Statistics

376. What is called the ratio of the dependent population to the earning population?
- (a) Dependency ratio (b) Sex ratio (c) Population density (d) Growth rate

377. Which of the following best describes the dependency ratio?

- (a) The ratio of the elderly population to the working-age population
- (b) The ratio of the combined non-working (0-14 and 65+) population to the working-age (15-64) population
- (c) The proportion of young dependents (0-14) in the population
- (d) The total population divided by the number of children (0-14)

378. City A has 12,000 individuals aged 0-14, 35,000 aged 15-64, and 5,000 aged 65+. What is the dependency ratio?

- (a) 0.31
- (b) 0.48
- (c) 0.60
- (d) 0.25

379. A City has a dependency ratio of 0.52. If its working-age population (15-64) is 50,000, what is the total number of dependents (0-14 and 65+)?

- (a) 15,600
- (b) 20,000
- (c) 26,000
- (d) 30,000

Answer the following 2 questions based on the information given below.

City	Population (in thousands)	Area (in km ²)
Gamma	1200	400
Delta	800	320

380. What is the population density of City Delta?

- (a) 2 people/km²
- (b) 4 people/km²
- (c) 2.5 people/km²
- (d) 2.2 people/km²

381. Which city is less densely populated?

- (a) Gamma
- (b) Delta
- (c) Both are equal
- (d) Cannot be determined

Answer the following two questions based on the information given below.

In a city, the total number of live births in a year was 2,400. The number of women aged 15-49 years in the population was 48,000.

382. Calculate the General Fertility Rate (GFR) for the city.

- (a) 40 per 1,000 women
- (b) 50 per 1,000 women
- (c) 60 per 1,000 women
- (d) 30 per 1,000 women

383. If live births increase to 3,000 while the number of women aged 15-49 remains the same, what is the new GFR?

- (a) 55 per 1,000 women
- (b) 65 per 1,000 women
- (c) 50 per 1,000 women
- (d) 62.5 per 1,000 women

384. The population of a city is 500,000, and the number of live births recorded in a year is 8,000. What is the Crude Birth Rate (CBR)?

- (a) 12 per 1,000
- (b) 16 per 1,000
- (c) 20 per 1,000
- (d) 22 per 1,000

385. What is the formula of population density?

- (a) $\frac{M}{F} \times 100$
- (b) $\frac{F}{M} \times 100$
- (c) $\frac{B}{P} \times 100$
- (d) $\frac{P}{A}$

386. In the following data, what is the dependency ratio?

Age	0-14	15-24	25-34	35-44	45-54	55-64	65+
Populataion	31,500	40,000	48,000	41,000	32,000	25,000	16,000

- (a) 35.54%
- (b) 25.54%
- (c) 23.24%
- (d) 31.25%

387. **Crude Birth Rate (CBR) is:**

- (a) $\frac{B}{P} \times 100$ (b) $\frac{B}{P} \times 1000$ (c) $\frac{P}{B} \times 100$ (d) $\frac{F}{P} \times 100$

388. **Which one is a measure of reproduction?**

- i) CBR
ii) CDR
iii) NRR

- (a) i (b) ii (c) iii (d) i and ii

389. **The number of people living per unit area is called–**

- (a) Population Index (b) Population Density
(c) Human Development Index (d) Dependency Ratio

390. **Which formula of GFR is accurate?**

- (a) $GFR = \frac{B}{P} \times 1000$ (b) $GFR = \frac{B}{F_{15-49}} \times 1000$
(c) $GFR = \frac{B_i}{F_i} \times 1000$ (d) $GFR = \frac{G_i}{F_{15-49}} \times 1000$

391. **Total number of children born to each 1000 people in any country or region is called –**

- (a) TFR (b) GFR (c) CBR (d) GRR

392. **A city has a dependency ratio of 0.48. If the working-age population (15–64) is 62,500, what is the number of dependents (ages 0–14 and 65+)?**

- (a) 30,000 (b) 25,000 (c) 22,000 (d) 20,000

393. **The dependency ratio of a town is 0.60. If there are 40,000 people aged 15–64, how many individuals are considered dependents?**

- (a) 22,000 (b) 26,500 (c) 24,000 (d) 25,000

394. **Which one is correct?**

- (a) $TFR_i GRR_i NRR$ (b) $TFR = GRR = NRR$ (c) $TFR_i GRR_i NRR$ (d) $NRR_i GRR_i TFR$

395. **Which reflectes an unchanging population?**

- (a) $GRR=1$ (b) $NRR_i 1$ (c) $GRR_i 1$ (d) $GRR=NRR$

9.1 Growth Rates

396. **If n in $P_n = P_o(1+r)^n$ is split into infinite parts and r adjusted accordingly, what type of growth do we have?**

- (a) Simple growth (b) Arithmetic growth (c) Exponential growth (d) Geometric growth

Answer the next two questions based on the following information

Year	1	2	3	4
Population	100	110	120	130

397. **Which type of growth is seen here?**

- (a) Arithmetic growth (b) Geometric growth (c) Exponential growth (d) None

398. **What is the rate of increase?**

- (a) 1 (b) 0.1 (c) 10 (d) 1%

399. In exponential growth, when is a population doubled?

- (a) $\frac{\log_{10} 2}{r}$ (b) $\frac{\log_e 2}{r}$ (c) $\frac{\log_e 2}{r^2}$ (d) $\frac{\log_e 3}{r}$

400. If a population exponentially declines, when is it reduced to half?

- (a) $\frac{\log_{10} 2}{r}$ (b) $\frac{\log_e 2}{r}$ (c) $\frac{\log_e 2}{r^2}$ (d) $\frac{\log_e 3}{r}$

401. How long does it take for a population to triple in exponential growth?

- (a) $\frac{\log_{10} 3}{r}$ (b) $\frac{\log_e 3}{r}$ (c) $\frac{\log_e 3}{r^2}$ (d) $\frac{\log_e 2}{r}$

Answer the next two questions based on the following information

Statement

402. Vital statistics records –

- i. marriage
- ii. birth
- iii. sickness and death

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

403. A population of 5000 grows at a geometric rate of 3% per annum. What will be the population after 6 years?

- (a) $5000 \times (1.03)^5$ (b) $5000 \times (1.03)^6$
(c) $5000 \times 6 \times 1.03$ (d) $5000 + 5000 \times 0.03 \times 6$

404. A population grows arithmetically from 10,000 to 12,000 in 5 years. What is the annual absolute increase?

- (a) 200 (b) 400 (c) 600 (d) 800

405. A bacterial culture grows exponentially with a growth rate constant of $r = 0.2$ per hour. How many hours will it take to triple?

- (a) $\frac{\ln 3}{0.2}$ (b) $\frac{\ln 2}{0.2}$ (c) $0.2 \times \ln 3$ (d) $0.2 \times \ln 2$

Answer Key:

1. (c) 6
2. (b) 6
3. (a) $\frac{n!}{(n-r)!}$
4. (a) $\frac{n!}{(n-1)!(n+r)!}$
5. (c) 1
6. (b) 0
7. (b) 2
8. (a) Disjoint
9. (b) Independent
10. (a) $P = \frac{\text{No. of favorable outcomes}}{\text{Total no. of possible outcomes}}$
11. (a) $P = \frac{\text{No. of favorable outcomes}}{\text{Total no. of possible outcomes}}$
12. (a) $P(A|B) = \frac{P(A \cap B)}{P(B|A)}$
13. (c) $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^{\infty} P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots$
14. (c) ii and iii
15. (b) Experiment
16. (c) An impossible event
17. (a) Complementary events
18. (c) 0
19. (d) Sample Point
20. (c) Mutually Exclusive Events
21. (a) $P(A \cap B) = P(A) \cdot P(B)$
22. (a) $A = \{1, 2, 3\}, B = \{4, 5\}$
23. (d) $P = \{m, n\}, Q = \{p, q\}$
24. (c) ii and iii
25. (a) i and iii
26. (a) i and ii
27. (d) All of the above
28. (d) $\frac{1}{2}$
29. (c) 0.90
30. (b) 0.85
31. (b) 0.92
32. (c) 0.88
33. (c) 0.975
34. (c) 6^r
35. (c) 2^r
36. (c) 8
37. (a) $\frac{1}{4}$
38. (c) $\frac{3}{4}$
39. (b) $\frac{1}{2}$
40. (a) $\frac{1}{4}$
41. (c) $\frac{3}{4}$
42. (c) $\frac{1}{4}$
43. (a) $\frac{3}{4}$
44. (a) An experiment
45. (b) $\frac{0}{6}$
46. (a) 4
47. (c) 36
48. (b) $\frac{1}{22}$
49. (b) $\frac{6}{11}$
50. (b) $\frac{1}{3}$
51. (b) $\frac{5}{12}$
52. (c) $\frac{48}{91}$
53. (d) $\frac{5}{11}$
54. (d) 0.0769
55. (c) $\frac{3}{4}$
56. (d) $\frac{3}{4}$
57. (b) 0.0769
58. (c) $\frac{3}{4}$
59. (a) $\frac{1}{2}$
60. (a) 0.26
61. (a) $\frac{13}{20}$
62. (c) $\frac{3}{4}$
63. (b) $\frac{4}{5}$
64. (c) $\frac{3}{4}$
65. (b) $\frac{9}{20}$
66. (a) $\frac{1}{8}$
67. (a) $P(A \cap B) = P(A) \times P(B)$
68. (b) $P(A \cup B) = P(A) + P(B)$
69. (d) $\frac{P(A)}{P(B)} = \frac{P(A|B)}{P(B|A)}$
70. (b) $\frac{1}{36}$

71. (a) $P(A) = 0.5$ and $P(B) = 0.4$ 96. (a) Probability distribution 121. (c) $\frac{5}{6}$
72. (a) 0.2 97. (b) 3 122. (a) $\frac{1}{3}$
73. (b) 0.4 98. (d) 3 123. (b) $\frac{1}{2}$
74. (c) 0.7 99. (d) 4 124. (b) $\frac{1}{2}$
75. (d) 0.1 100. (c) $\int_x f(x) dx = 1$ 125. (c) $\frac{1}{4}$
76. (c) 0.7 101. (d) Size of television 126. (c) $\frac{5}{6}$
77. (a) 0.1 102. (b) $\frac{1}{20}x^5 + c$ 127. (a) $\frac{1}{2}$
78. (c) ii and iii 103. (b) i and iii 128. (c) $\frac{1}{20}$
79. (c) $\frac{1}{4}$ 104. (d) i, ii, and iii 129. (a) $\frac{5}{12}$
80. (a) $\frac{1}{4}$ 105. (d) i, ii, and iii 130. (c) $\frac{1}{2}$
81. (d) $\frac{11}{12}$ 106. (d) 1 131. (c) $\frac{1}{2}$
82. (b) $\frac{1}{4}$ 107. (c) 0 132. (d) $\frac{1}{6}$
83. (c) $\frac{3}{20}$ 108. (a) 2 133. (a) $\frac{5}{6}$
84. (a) $\frac{17}{20}$ 109. (b) Weight 134. (b) $\frac{5}{6}$
85. (c) $\frac{58}{60}$ 110. (c) $\sum P(x_i) = 1$ 135. (c) $\frac{1}{2}$
86. (b) $\frac{7}{30}$ 111. (d) Released version number of a software 136. (d) $\frac{5}{5}$
87. (d) $\frac{23}{30}$ 112. (b) Weight of a package 137. (c) 1
88. (a) $\frac{3}{7}$ 113. (c) Number of floors in a building 138. (c) $\frac{1}{21}$
89. (d) 0.4 114. (c) Number of defective items in a batch 139. (b) Discrete random
90. (c) ii and iii 115. (b) Height of students 140. (c) 14
91. (c) ii and iii 116. (d) $0 \leq P(X_i, Y_j) \leq 1$ 141. (c) $\frac{5}{14}$
92. (b) i and iii 117. (b) 0.75 142. (c) Probability mass function
93. (a) i and ii 118. (a) 0.75 143. (c) 1
94. (b) i and iii 119. (c) $\frac{2}{3}$ 144. (c) 1
95. (d) Page size in word processing software 120. (d) $\frac{1}{2}$ 145. (d) 1

146. (c) 1
147. (a) $P(x) = \frac{2x+3}{21}$
148. (c) $\frac{3y+2}{7}$
149. (b) $\frac{1}{6}$
150. (a) $\frac{5}{6}$
151. (a) i
152. (a) i and ii
153. (d) i, ii and iii
154. (d) i, ii and iii
155. (d) i, ii and iii
156. (d) i, ii and iii
157. (c) ii
158. (b) $E(X) + E(Y)$
159. (b) $4E(X) + 2E(Y)$
160. (b) 2
161. (d) 8
162. (c) Variance
163. (c) 0
164. (b) $a^2V(X)$
165. (a) $aE(X) + b$
166. (a) 0
167. (c) $\frac{n+1}{2}$
168. (c) 16
169. (c) a
170. (a) 0
171. (c) $V(X) - V(Y)$
172. (c) $4V(X)$
173. (a) 1
174. (d) Central tendency
175. (c) 2
176. (b) 5.5
177. (b) $V(X + Y) = V(X) + V(Y) + 2Cov(X, Y)$
178. (b) i
179. (c) 4
180. (a) 2
181. (a) 10
182. (b) 5
183. (a) 4
184. (b) 6
185. (c) 31
186. (a) 2
187. (c) 21
188. (c) 16
189. (b) 9
190. (a) 10
191. (c) 48
192. (d) 5
193. (d) 36
194. (b) 9
195. (d) 0
196. (b) 5
197. (a) 20
198. (c) 4
199. (a) 3
200. (b) 6
201. (c) 4
202. (b) $2\sqrt{6}$
203. (c) 4
204. (a) 1
205. (b) 5
206. (a) 1
207. (c) -1
208. (d) 32
209. (d) 8
210. (b) $\frac{13}{12}$
211. (b) $\frac{13}{12}$
212. (d) 2.97
213. (b) 1.9
214. (b) 1.29
215. (a) 3.6
216. (b) 3.36
217. (b) 3.4
218. (b) 2.44
219. (c) 3.6
220. (a) 2.24
221. (c) 3.0
222. (c) 1.00

223. (c) 2.33
224. (b) 0.56
225. (b) 2.17
226. (a) 5.17
227. (d) 4.25
228. (b) 10
229. (d) 3.0
230. (d) 24
231. (b) 2.92
232. (a) 1.05
233. (c) 10
234. (c) 0.725
235. (d) i, ii and iii
236. (b) 2
237. (a) $Mean > Variance$
238. (a) np
239. (b) npq
240. (b) npq
241. (d) $\sqrt{npq}$
242. (d) $\sqrt{\frac{q}{np}}$
243. (c) $np > 0$
244. (b) Mean is greater than variance
245. (c) Mean is greater than variance
246. (a) 2
247. (a) $P(x+1) = \frac{n-x}{x+1} \frac{p}{q} P(x); x = 0, 1, 2, \dots, n$
248. (a) $\beta_1 = \frac{(q-p)^2}{npq}$
249. (a) $\beta_2 = 3 + \frac{1-6pq}{npq}$
250. (d) $p = q$
251. (a) $n \rightarrow \infty$ and $p \rightarrow 0$
252. (d) $n \rightarrow \infty$ and $p \cong q$
253. (b) $p \neq q$
254. (a) $p \neq q$
255. (a) Binomial variate
256. (c) 0.3828
257. (c) 0.1419
258. (a) 0.2734
259. (c) 0.1032
260. (b) 0.2508
261. (b) 10
262. (c) 12
263. (c) 8
264. (d) 15
265. (b) 9
266. (a) $16, \frac{1}{4}$
267. (c) 0.99
268. (d) 6.25
269. (b) 0.0069
270. (a) i and ii
271. (c) 0.80
272. (b) 0.6723
273. (a) Positive skewness and leptokurtic
274. (a) 0
275. (a) i and ii
276. (a) 28
277. (d) Negative skewness and leptokurtic
278. (d) 0.7037
279. (d) Negative skewness and leptokurtic
280. (c) 400
281. (d) 0.1
282. (b) 0.54
283. (c) 0.140
284. (b) 0.103
285. (a) 8.75
286. (a) 3
287. (b) 5.69
288. (a) 2.25
289. (a) 2.38
290. (a) 1.60
291. (a) 0.02
292. (a) 0.19
293. (c) 2.94
294. (c) 3.09
295. (a) mean $\neq$ variance
296. (a) mean $\neq$ variance
297. (b) Positive skewness and platykurtic
298. (a) Positive skewness and leptokurtic
299. (b) i and iii

300. (a) i and ii
301. (b) i and iii
302. (d) i, ii and iii
303. (d) i, ii and iii
304. (d) i, ii and iii
305. (c) i and iii
306. (b) ii and iii
307. (c) i and iii
308. (a) i and ii
309. (a) i and ii
310. (a) i and ii
311. (d) i, ii and iii
312. (a) i and ii
313. (d) i, ii and iii
314. (d) i, ii and iii
315. (d) i, ii and iii
316. (a) i and ii
317. (a) 1
318. (b) m
319. (c) $Mean = Variance$
320. (c) m
321. (c) $\sqrt{m}$
322. (c) $m > 0$
323. (b) 5
324. (a) $n \rightarrow \infty, p \rightarrow 0$ & np is finite
325. (d) 2
326. (b) 5
327. (b) 3
328. (b) 3.46
329. (a) 4.5
330. (c) 2.45
331. (a) 3
332. (a) 1.978
333. (c) 0.15
334. (c) 4
335. (c) $a^{\frac{1}{2}}$
336. (b) 9
337. (a) 5
338. (a) 0.1465
339. (b) 0.1057
340. (c) 2.65
341. (b) 1.2515
342. (b) 0.1954
343. (a) 3
344. (d) 0.0024
345. (b) $\sqrt{10}$
346. (b) 8
347. (b) i and iii
348. (d) i, ii and iii
349. (b) i and iii
350. (d) i, ii and iii
351. (d) i, ii and iii
352. (a) 0.036
353. (d) $-\infty$ to ∞
354. (a) 0
355. (a) $(-\infty, \infty)$
356. (b) $Z = \frac{X-\mu}{\sigma}$
357. (c) 0
358. (c) 0
359. (b) $\frac{1}{\sigma\sqrt{2\pi}}$
360. (a) 0
361. (c) 2
362. (b) 0.56
363. (c) 0.997
364. (c) 180
365. (d) Current year
366. (d) Marshal-Edgeworth index number
367. (a) Laspeyres' index number
368. (b) Statistic
369. (c) Census
370. (c) Parameter
371. (a) Simple random sampling
372. (d) 10000
373. (c) 210
374. (a) $\frac{1}{N^n}$
375. (a) i and ii
376. (a) Dependency ratio

377. (b) The ratio of the combined 15-64 and 65+ population to the working-age population (15-64) population
378. (b) 0.48
379. (c) 26,000
380. (b) 4 people/km²
381. (b) Delta
382. (b) 50 per 1,000 women
383. (d) 62.5 per 1,000 women
384. (b) 16 per 1,000
385. (d) $\frac{P}{A}$
386. (b) 25.54%
387. (b) $\frac{B}{F_{15-49}} \times 1000$
388. (c) iii
389. (b) Population Density
390. (b) $GFR = \frac{B}{F_{15-49}} \times 1000$
391. (c) CBR
392. (a) 30,000
393. (c) 24,000
394. (a) TFR_iGRR_iNRR
395. (a) GRR=1
396. (c) Exponential growth
397. (a) Arithmetic growth
398. (b) 0.1
399. (b) $\frac{\log_e 2}{r}$
400. (b) $\frac{\log_e 2}{r}$
401. (b) $\frac{\log_e 3}{r}$
402. (d) i, ii and iii
403. (b) $5000 \times (1.03)^6$
404. (b) 400
405. (a) $\frac{\ln 3}{0.2}$